Advanced Hybrid Deep Learning Approach Using Xception And Bi-Lstm For Early Detection Of Diabetic Retinopathy

By

Pooja Pachipala

Student ID: 20064859

MSc in Data Analytics

Supervisor: Prof. Muhammad Asad

Declaration

I declare that all the work in this dissertation is entirely my own unless the words have been placed in inverted commas and referenced with the original source. Furthermore, texts cited are referenced as such, and placed in the reference section. A full reference section is included within this thesis.

No part of this work has been previously submitted for assessment, in any form, either at Dublin Business School or any other institution.

Signed: Pooja Pachipala

Date: 11/05/2026

Acknowledgements

I would like to thank my family for their support during the last year, without their support the completion of this program and thesis would not have been possible. Also I wish to express my gratitude to my supervisor Muhammad Asad, without his support and creative impulses this thesis would not be as it is today.

Abstract

Diabetic retinopathy detection is an important area of medical image analysis that aims to support early diagnosis and reduce the risk of vision loss among diabetic patients. The aim of this study is to build an accurate and reliable automated DR classification system that can assist in the better grading of DR and aid in rapid clinical decision making. Existing diabetic retinopathy detection approaches, including traditional machine learning and convolutional neural network-based methods, mainly focus on spatial feature extraction while showing limitations in sequential lesion modelling, class imbalance handling, limited generalization, and reduced robustness across datasets and severity levels. To overcome these drawbacks, this study introduces a novel hybrid model called Xception-BiLSTM, which integrates efficient spatial feature extraction with a bidirectional sequential network to model the relationship and context among lesions. The proposed approach involves image preprocessing, contrast enhancement, augmentation, and imbalanced data handling to enhance the quality of features and model stability. The study utilizes the IDRiD dataset, which consists of retinal fundus photographs with diabetic retinopathy severity labels, for data modeling and assessment. The implementation is conduct on GPU-based places like Google Colab and Kaggle, using Python, TensorFlow, Keras, OpenCV, NumPy, and Pandas to boost computational efficiency. The proposed model had moderate validation accuracy of 68.75% while achieving moderate precision, recall, and F1-score across classes of diabetic retinopathy. The research is valuable for healthcare systems, doctors, and telemedicine platforms, as it facilitates automated DR screening at scale, with reliability and efficiency.

Keywords: Diabetic Retinopathy Detection, Hybrid Deep Learning, Xception–BiLSTM, Retinal Fundus Image Classification, Automated Medical Screening

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CHAPTER 1: INTRODUCTION

1.1 Chapter Introduction

This chapter discusses DR as a global health problem and as a major contributor to avoidable blindness in the working adult population. The risk of developing retinal complications and losing sight is also growing as the number of cases of diabetes is growing all over the world. It is crucial to detect it as early as possible, through regular screening, to prevent any permanent damage; however, this is still challenging to do on a large scale manually. Hence, automated detection systems of DR should be provided to aid prompt diagnosis and minimize clinical workload. The detection methods have evolved from the traditional machine learning methods that expand the deep learning model with features of a hierarchical feature extraction. There are new hybrid models that combine spatial and contextual learning that have been developed recently. This chapter provides background information, a problem statement, research goals, rationale, contributions, and dissertation organization.

1.2 Background of the Study

DR is one of the leading causes of world vision loss and must be screened for in early stages. Since diabetes has spread throughout the world, vision-threatening complications have been on the rise (Kropp et al., 2023). In recent years, WHO has estimated several times the burden of diabetes-related visual impairment, particularly in developing and middle-income countries where access to eye care is limited. (Das *et al.*, 2024). Timely retinal screening can identify the disease early enough and can save serious cases of permanent blindness (Hassan et al., 2024). Thus, not only is it a clinical need to increase DR screening programs, but also a critical public health concern at the international level.

While important, there are several difficulties with manual grading of the retinal fundus images. Very time consuming, especially in large scale screening operations where thousands of images have to be analysed (Zedan *et al.*, 2023). Proper diagnosis needs qualified and skilled ophthalmologists, which are not always available in a resource-starved system. Moreover, inter-observer variability also brings subjectivity, as the judgment of a specialist can be different among specialists depending on the experience and interpretation. These inconsistencies can affect the diagnostic reliability and delay the treatment decisions. The restrictions indicate that more frequent, improved, and standardized ways of detecting diabetic retinopathy are required (Chauhan et al., 2025).

Medical image analysis and automated disease detection have seen a significant revolution with the advent of AI, especially when using CNNs for retinal image classification. CNN-based models are able to learn the hierarchical spatial features and attain impressive DR grading results. Nevertheless, generalization can be a problem with pure spatial deep learning models as they tend to not predict contextual progression of lesions. Also, the implementation into the actual clinical practice requires computationally efficient, reliable and scalable system. Hence, it is essential to come up with sophisticated automated screening systems to address increasing healthcare needs efficiently and fairly across the globe.

1.3 Problem Statement

The existing diabetic retinopathy detection machines have high performance, but still have significant limitations. Majority of methods are based on spatial CNN designs that focus on feature extraction, but ignore the dynamics of lesion progression. For instance, Madan et al. (2022) proposed a CNN-BiLSTM model for diabetes diagnosis without detailed spatial-sequential fundus grading. On the same note, Wang et al. (2025) suggested an attention-enhanced residual CNN, which enhanced spatial discrimination but did not have an effective progression modeling. Alavee et al. (2024) fused explainable AI with the Xception, gaining some interpretability and suffering from the drawbacks of overfitting and cross-device validation. Also, incessant imbalance of the classes between the mild and severe cases, high computational complexity and lack of cross-dataset robustness diminish real-world deployability. Lack of entirely incorporated spatial-sequential framework limits degrees of grading credibility and scaling. Thus, the paper fills these gaps by integrating the spatial efficiency of Xception and BiLSTM-based sequential modelling to make the grading robust on the IDRiD dataset.

1.4 Research Aim & Research Objectives

To design an enhanced Hybrid Xception–BiLSTM DL model to improve the accuracy and reliability of diabetic retinopathy detection using the IDRiD dataset.

Research Objectives

1. To develop a hybrid model integrating Xception for spatial feature extraction and Bi-LSTM for sequential lesion learning.
2. To apply preprocessing, augmentation, and segmentation techniques to enhance fundus image quality and model performance.
3. To evaluate the hybrid model against existing deep learning baselines for accuracy, sensitivity, and robustness.

1.5 Research Question

How effectively can the proposed Xception–BiLSTM hybrid model improve diabetic retinopathy classification performance on the IDRiD dataset compared to existing deep learning methods?

1.6 Research Rationale

Classical convolutional neural network models are based on spatial features recognition to detect lesions in retinal fundus images, including microaneurysms, hemorrhages, and exudates. Although they are useful in identifying visible abnormalities, they cannot identify lesion distribution, density, and progression which are critical in determining the severity of diabetic retinopathy. Spatial architectures are frequently unable to capture transition between disease stages. Sequential and contextual modeling is important clinically, particularly in the differentiation between mild and moderate non-proliferative DR. Scalable and robust models are required for large-scale screening across datasets and devices. This paper is a combination of Xception to extract spatial features efficiently and BiLSTM to learn contextually in both directions based on IDRiD dataset.

1.7 Research Contribution

1. Established the primary goal of this research as improving the accuracy, reliability, and robustness of automated diabetic retinopathy grading through integrated spatial–sequential deep learning modeling for clinically meaningful screening support.
2. Developed and evaluated a hybrid Xception–BiLSTM architecture for automated diabetic retinopathy classification, demonstrating reliable classification performance and contextual lesion learning capability.

3. Utilized the IDRiD dataset to ensure structured lesion-level evaluation, incorporating preprocessing, augmentation, and imbalance-handling strategies to enhance generalization and grading performance.
4. Achieved a validation accuracy of 68.75% with improved grading robustness and reliable classification performance compared with baseline deep learning approaches.

1.8 Structure of the Dissertation

The research is presented in 5 chapters systematically. The study is introduced in Chapter 1 which covers background on DR, problem statement, aim, objectives, and rationale. Chapter 2 provides an overview of the previous ML, DL, and hybrid DR detection models and gaps in research. The methodology is explained in Chapter 3, encompassing the IDRiD dataset, preprocessing, suggested Xception–BiLSTM architecture, and training strategy. The experimental results, evaluation, and comparative analysis are given in Chapter 4. The results, conclusions, constraints, and recommendations for future study are presented in Chapter 5.

1.9 Chapter summary

The chapter describes diabetic retinopathy as an important health problem and a preventable cause of blindness, especially for those who have chronic diabetes. The rising number of cases and shortage of specialists highlight the need for automated screening systems. The development of DR detection through old machine learning in which features are handcrafted to modern deep learning and hybrid features is addressed. In spite of this progress, there are still gaps such as deficiency of spatial-sequential integration, class imbalance, inability to cross-dataset validate, and scalability. In order to curb these, a hybrid Xception-BiLSTM model is suggested to enhance the reliability and robustness of grading. The literature review is presented in detail in the next chapter.

CHAPTER 2 LITERATURE REVIEW

2.1 Chapter Introduction

This chapter gives a critical review of the existing DR detection and classification literature. It discusses the clinical background of DR, classical machine learning classifiers, CNN based models, transfer learning based models, attention models, and hybrid deep learning models. Current methodologies are critically evaluated in terms of their advantages and limitations, computational trade-offs, interpretability, scalability and generalization to real-world screening applications. Furthermore, the chapter describes the progress in DR detection from basic machine learning methods to more sophisticated deep learning and hybrid methods, highlighting key developments, problems and opportunities in the current literature. The review also highlights significant research gaps, such as limitations in lesion representation, dataset generalization and model robustness, that need to be explored in automated DR detection systems.

2.2 Overview of Diabetic Retinopathy

DR is a serious complication of diabetes known as a microvascular complication, affecting the blood vessels of the retina and is potentially sight-threatening if left untreated. It is usually divided into non-proliferative and proliferative phases, becoming more severe in terms of microaneurysms, hemorrhages, exudates and neovascularization. Regular retinal screening is very crucial in the prevention of blindness. Nevertheless, automated assessment of fundus images is tedious and manual, which proves the necessity to present effective automated diagnostic tools to assist the clinical decision-making process and screening networks.

2.2.1 Clinical Background and Disease Progression

Sun et al. (2023) give the elaboration of the molecular mechanism of initial pathological processes of diabetic retinopathy of the involvement of oxidative stress, inflammation, endothelial dysfunction, and blood-retinal barrier disintegration. Although biologically informative and highlighting the importance of early detection of NPDR, this work is again primarily pathophysiological and does not incorporate computational imaging or AI approaches for detection.

The study conducted by Alanazi and Alanazi (2025) is on the development of federated CNN for the detection of diabetic retinopathy progression. Although model is more scalable and has better privacy, it does not provide lesion-level explanations or pathophysiological explanations. Furthermore, The model may be expensive to compute and not appropriate for screening applications.

Serdiuk et al., (2024) outlined the treatment intervention strategies for various stages of DR in type 2 diabetes, and linked biomarkers to the risk of progression. The authors carefully distinguish NPDR, characterized by microaneurysms and hemorrhages, from PDR, characterized by neovascularization and the risk of retinal detachment. Although the article is devoted to the effectiveness of treatment, it also covers statistical modeling of prognosis, which is of interest for AI-based grading. However, the lack of CNN or deep learning methods for quantifying lesions is a limitation to automation. As illustrated in Figure 1, DR is a significant cause of blindness globally, a fact highlighted by the World Health Organization.

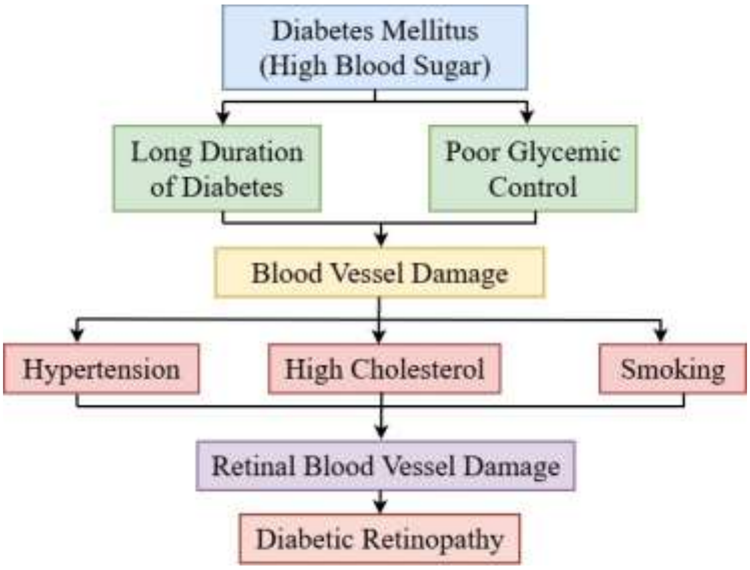

Figure 1: Causes and Progression of Diabetic Retinopathy

2.2.2 Clinical Challenges in Early Detection

Abou Taha et al., (2024) critically analyze economic and systemic issues in country-wide screening for diabetic retinopathy, pointing out the issues of human resource shortages, inter-observer variability, and resource-intensive manual grading. Although a very strong piece of work from a public health point of view, the narrative review does not contain any empirical analysis of AI systems.

The article by Grauslund (2022) talks about the screening of DR in the age of AI, focusing on both the positive aspects and the challenges of deep learning algorithms. Although the article focuses on the aspects of interpretability, accountability, and regulation in the clinical setting, it is conceptual in nature and does not provide any new experimental evidence.

In a critique of the use of AI to screen for DR using colour fundus images, Grzybowski et al. (2023) highlight the limitations of manual grading, which include inter-observer variability and subjective identification of lesions. The article speaks about CNN based screening systems with special focus on governance, validation and integration. However, bias in the data, absence of external validation and infrastructure demands hinder scalability, even in situations where the workload of ophthalmologists is less, as fewer ophthalmologists are needed.

2.3 Traditional Machine Learning Approaches in DR Detection

The conventional machine learning methods for DR detection mostly rely in manually designed feature extraction methods and generally used classifiers like SVM, K-NN and RF. The manually designed features include the texture, morphological and histogram-based features that strive to feature the lesion using retinal fundus images. These methods are computationally efficient, too, and are comparatively interpretable, yet their performance highly depends on feature engineering. Poor scalability, inability to adapt to new data and learn hierarchical representations are weaknesses, which restrict their applicability in large scale clinical screening applications. Table 1 shows the summary of Conventional Machine Learning Methods in DR Detection.

2.3.1 Feature Engineering-Based Models

In the work of Usman et al. (2023), Principal Component Analysis was used with manually designed texture, morphological, and histogram-based features to detect diabetic retinopathy, and the result was an improvement in the reduction of dimensionality and computational complexity. While PCA helped in improving the robustness of features and reducing redundancy, the method was still highly reliant on manually designed features.

Sambasiva Rao and Pande (2024) proposed a hybrid approach utilizing a hand-crafted texture and morphological features with MobileNet for fundus disease classification. The combination enhanced the performance of lightweight CNNs but still remained vulnerable to illumination, resolution, and device variations. The approach is limited in its ability to scale because there was not much experimental work nor there were domain-specific feature engineering.

Bogacsovics et al. (2022) proposed a hybrid CNN model that used handcrafted texture, morphological and histogram features together with deep features extracted from pre-trained CNN models for DR classification. By combining hand-engineered descriptors with feature maps extracted from CNNs, the proposed architecture improves the identification of lesions, especially for small datasets. Nevertheless, the combination of features results in increased dimensionality, complexity, and the potential for overfitting. The use of hand-engineered features is less flexible and scalable, particularly in real-world screening settings involving various imaging devices.

2.3.2 Classical Classifiers

Naveen et al. (2025) proposed the EffNet-SVM model, in which EfficientNetV2-Small derives strong features, and SVM is used for the final DR grading. Combination of the two models enhances class separability using margin optimization and performs better than a CNN softmax classifier. Nevertheless, the two-stage model restricts end-to-end learning, is computationally intensive for large datasets, and could be less robust due to the SVM kernel’s sensitivity to heterogeneous retinal images.

Ullah et al. (2022) suggested a Genetic Algorithm based feature optimization framework coupled with CNN and ECOC-SVM based diabetic retinopathy multi-classifier. The proposed algorithm improves feature selection and class separation by using evolutionary optimization and error-correcting features. Nevertheless, real time scaling is restricted by increased computational cost, longer training time and sensoritivity of kernels.

Bhimavarapu et al. (2024) compared feature extraction and SVM and KNN classifiers with enhanced CNN, demonstrating optimized SVM, which has good results in cross-validation. The results indicate that traditional classifiers can be competitive provided that they possess high quality discriminative features. These methods are feature-based, are not based on hierarchical spatial relationships, and do not necessarily transfer to other imaging devices and other real-world settings as listed in Table 1.

Table 1: Summary of Traditional ML Approaches in DR Detection

Author(s)	Methods	Advantages	Limitations
Naveen et al. (2025)	EfficientNetV2-Small + SVM (EffNet-SVM)	Enhanced class separability; better than CNN softmax classifier	Restricts end-to-end learning; computationally intensive; SVM kernel sensitivity affects robustness
Bhimavarapu et al. (2024)	CNN-based feature extraction + SVM/KNN	Competitive performance with optimized SVM; effective cross-validation results	Feature-dependent; lacks hierarchical spatial modeling; weak generalization across devices
Sambasiva Rao & Pande (2024)	MobileNet combined with hand-engineered texture and morphological features	Enhanced performance of lightweight CNN; improved feature integration	Sensitive to illumination, resolution, and device variation; limited scalability and domain-specific feature engineering
Usman et al. (2023)	PCA with manually designed texture, morphological, and histogram features	Reduced dimensionality and computational complexity; improved feature robustness and redundancy reduction	Highly dependent on hand-crafted features; limited adaptability
Ullah et al. (2022)	Genetic Algorithm + CNN + ECOC-SVM	Improved feature selection and class separation; evolutionary optimization benefits	High computational cost; increased training time; limited scalability
Bogacsovics et al. (2022)	Hybrid CNN with hand-engineered descriptors + deep features	Improved lesion detection, especially for small datasets	Increased dimensionality and complexity; risk of overfitting; limited flexibility in real-world screening

2.4 Deep Learning in DR Detection

DL has contributed immensely in the detection of DR by providing the ability to learn hierarchical features in a retinal fundus directly through the images. CNNs have been reported to be better than conventional methods in their ability to detect lesions and classify severity based on multi-classes. Other architectures like ResNet, Inception, and EfficientNet improve the quality of spatial features extraction and classification. Transfer learning also enhances generalization using small medical data. In spite of these advances, such issues as class imbalance, dataset bias, lack of interpretability, and high computational complexity still pose a problem to robustness and real-world clinical application. Table 2 shows the summary of DL in Diabetic Retinopathy Detection.

2.4.1 Convolutional Neural Networks in Medical Imaging

In this study, Asia et al. (2022) evaluated the performance of different CNN architectures for automatic detection of DR from retinal fundus images and demonstrated their effectiveness in automatic feature learning and spatial learning at various levels. The proposed approach could provide automated detection of microaneurysms, hemorrhages and exudates without the need for hand engineered features. Nonetheless, work was focused on dataset-specific validation, had poor generalization across datasets, and did not provide a thorough analysis of interpretability.

A text-based CNN architecture with pre-data texture information is proposed in the works of Ishtiaq et al. (2023) to enhance DR classification using InceptionV3 based on the ensemble optimization method, DenseNet121 and EfficientNet-B0 architectures. The approach improved toughness and localization of lesions of different sizes. Nonetheless, the handcrafted features added cost to the computation and excessive complexity in architectures restricted real-time processing in large-scale screening settings.

Prabhavathy et al. (2022) compared the performance of pretrained CNN models such as InceptionV3, DenseNet121, EfficientNet-B0 with other traditional ML models and found it to be better on APTOS and Kaggle DR datasets. The models allowed progressively learning lesions using factorized convolutions, dense connections and compound scaling. Although multi-class grading showed high performance, the limitations were data-specific training, class imbalance, limited cross-validation, low interpretability and large annotated datasets.

2.4.2 Transfer Learning Approaches

In their study, Kallel and Echtioui (2024) examined application of TL in diabetic retinopathy regarding classification through fine-tuning CNNs models trained on retinal fundus. They employed the preprocessing, data augmentation, and layer-wise fine-tuning strategy which allowed them to converge faster and achieve greater accuracy than training with random initialisation. Nevertheless, the domain differences between natural and retinal images can cause the decreased relevance of deep features and the increased risk of overfitting, in case of a severe DR.

Jabbar et al. (2022) suggested that the framework of transfer learning to diagnose diabetic retinopathy is based on weighted loss functions and pretrained CNN models with image resizing. Deep layer fine-tuning enhanced the discrimination between neighbouring levels of severity whereas EfficientNet optimized the extraction of features. Even though it performed well on small datasets, robustness was impacted by small domain gaps across cross-devices and the ImageNet domain.

Nagpal et al. (2023) have proposed a transfer learning model to classify DR and HYBR from medical images, where ImageNet-pretrained CNN models are used, with fine-tuning methods, such as either fully connected layers replacement or selective convolutional layer unfreezing to enhance the adaptation. The proposed framework improved the training time, computational complexity, and required data without compromising the accuracy. Nonetheless, small datasets could be prone to overfitting, lack of control over class imbalances and domain gaps between fundus and natural images can impose restrictions on clinical generalizability and reliability.

2.4.3 Performance Comparison and Limitations

Dai et al. (2024) introduced DeepDR Plus, a CNN-based approach that forecasts the progression of DR instead of merely classification. The proposed model represents biomarkers of vascular and lesion changes to predict individual risk timelines. In spite of the effectiveness of this model, a problem of population bias, unbalanced classes, interpretation, and high computations is a critical issue.

A new system of automated deep learning screening by Guefrachi et al. (2024) that utilizes image preprocessing and CNN-based classification to improve lesion-level spatial representation was proposed. Even though the detection accuracy increased, assessment measures were influenced by class imbalance, especially underrepresented extreme cases. There was poor interpretability and too much computational complexity to use in practice.

Atwany et al. (2022) completed review of the developments in CNN, transfer learning, ensemble, and attention models to classify diabetic retinopathy using EyePACS, APTOS, and Messidor datasets. Although models like VGGNet, ResNet, Inception, DenseNet, and EfficientNet performed well in hierarchical feature extraction and accuracy, robustness to real-world challenges was still a concern. Biases in the dataset, class imbalance issues in the most severe form of PDR, interpretability, and computational complexity are some factors that impede scalability as Table 2.

Table 2:Summary of Deep Learning in Diabetic Retinopathy Detection

Author(s)	Methods	Advantages	Limitations
Guefrachi et al. (2024)	Preprocessing + CNN-based lesion classification	Improved lesion-level spatial representation; better detection accuracy	Severe class imbalance; lack of interpretability; computational complexity
Dai et al. (2024)	DeepDR Plus (CNN-based progression prediction)	Predicts DR progression timeline; biomarker-based modeling	Dataset bias (population-specific); imbalance; interpretability issues; high computational cost
Kallel & Echtioui (2024)	Transfer learning with ImageNet pretrained CNNs + fine-tuning	Faster convergence; improved accuracy over training from scratch	Domain mismatch; imbalance-related overfitting risk
Nagpal et al. (2023)	Transfer learning with selective layer unfreezing	Reduced training time; lower data requirement; maintained accuracy	Overfitting in small datasets; imbalance issues; domain mismatch limits generalizability
Ishtiaq et al. (2023)	Ensemble CNN (InceptionV3, DenseNet121, EfficientNet-B0) + texture features	Improved robustness and spatial abstraction for multi-size lesions	Increased computational cost; architectural complexity; limited real-time feasibility

Asia et al. (2022)	CNN architectures for automatic DR detection	Automatic hierarchical feature learning; effective lesion detection without handcrafted features	Dataset-specific validation; poor cross-dataset generalization; limited interpretability
Prabhavathy et al. (2022)	Pretrained CNNs (InceptionV3, DenseNet121, EfficientNet-B0) on APTOS & Kaggle	Strong multi-class grading; effective progressive lesion learning	Dataset-specific training; class imbalance; limited cross-validation; low interpretability; large data requirement
Jabbar et al. (2022)	Transfer learning + weighted loss + EfficientNet	Better severity-level discrimination; good small dataset performance	Limited cross-device validation; domain gap affects robustness
Atwany et al. (2022)	Review of CNN, TL, ensemble, attention models (EyePACS, APTOS, Messidor)	Strong hierarchical feature extraction; high accuracy	Dataset bias; imbalance (PDR); interpretability limits; scalability concerns

2.5 Hybrid Deep Learning Models

Hybrid deep learning networks combine two or more architectures in order to benefit and detect diabetic retinopathy through complementary spatial and contextual information. Convolutional Neural Networks are usually paired with recurrent networks e.g., LSTM or BiLSTM or augmented by attention to model the sequential dependence and improve the lesion representation. The objectives of these models are to attain a trade-off between contextual knowledge and feature extraction, which results in better classification. Nevertheless, greater complexity in architecture, and computational complexity, and possible overfitting are also leading issues that need to be optimized and verified.Table 3 shows the summary of Summary of Hybrid Deep Learning Models.

2.5.1 CNN + LSTM Architectures

Madan et al. (2022) suggested a CNN-BiLSTM architecture for real-time diabetes prediction. While not applicable to fundus grading, the architecture does show the integration of spatial and contextual information. Nevertheless, its use in structured, non-image data hinders its validation at the lesion level, and Bi-LSTM layers make it computationally complex, thereby hindering scalability for high-resolution retinal imaging.

A CNN-LSTM hybrid model was proposed by Khodadadi et al. (2025) for retinal OCT image classification, combining spatial feature learning with the analysis of context relationships. This method further improved the model's ability to capture small-scale structural features related to disease development. However, the use of sequential interpretation on static fundus images might decrease conceptual alignment, and the additional complexity of the model architecture would increase the risk of overfitting.

In addition, Kumar et al. (2024) developed a hybrid R-CNN-LSTM model for DR detection that combines region-based feature detection with context modeling using LSTMs. The R-CNN module is used for the localization of lesions in the retinal images, and the LSTMs are used to capture the relationships between microaneurysms, hemorrhages, and exudates. The model performs better in the fine-grained grading of diabetic retinopathy, especially in distinguishing between the early and moderate NPDR stages, compared to the baseline CNN models. Nevertheless, region-based recurrence leads to increased complexity and the use of feature sequences instead of actual longitudinal observations.

2.5.2 CNN + Attention and Transformer Models

A multi-attention residual refinement CNN was proposed by Wang et al. (2025), incorporating class-specific attention modules to focus on diagnostically important retinal areas. Residual learning with spatial attention enhanced discriminative power compared to vanilla CNNs. Nevertheless, additional complexity due to multiple attention layers and reliance on large-scale annotated datasets hinder real-time performance and stability in screening applications.

Miao and Tang (2022) proposed a multiscale hybrid attention-residual network with dual-attention modules based on the transformer idea to improve feature fusion at the lesion scale. While improved contextual modeling increased the accuracy of classification, the high computational complexity and memory requirements impede efficiency. Moreover, the susceptibility to large training datasets poses a risk of overfitting and hinders cross-device applications in MI.

Li et al. (2022) proposed an ECA-CBAM attention-enhanced CNN, consisting of channel and spatial attention, to enhance the extraction of lesion-centered features for the DR severity assessment. Cross-combined attention modules enhanced discriminative capabilities compared to traditional CNNs by focusing on diagnostically important areas. Although improved modelling of contextual information and alignment with transformer-based architectures has been achieved, the added parameter cost, computational requirements, and dependence on large-scale annotated datasets pose scalability issues.

2.5.3 Xception-Based Hybrid Models

Sathiya et al. (2024) compared InceptionV3 and Xception models for diabetic retinopathy classification tasks, showing that depthwise separable convolutions in Xception enhance parameter efficiency and spatial representation. Although the model has competitive accuracy for multi-class grading of high-resolution images of fundus, model is strictly spatial. The lack of sequential 24odelling makes it difficult to 24odelli the dependency between lesions, especially when there is overlap between mild and moderate DR.

Alavee et al. (2024) suggested an improved Xception-Inception model using Grad-CAM-based explainable AI for early detection of DR. While interpretability and accuracy of classification were improved, model was still spatially concentrated and lacked the ability to model sequential processes. Higher complexity in models can lead to overfitting in smaller datasets, thus emphasizing the importance of hybrid models that are both spatial and sequential.

Reddy et al. (2026) suggested a hybrid Xception-based deep learning model combined with explainable AI to classify diabetic retinopathy through the Use of depth wise separable convolutions to protect parameters and better localize lesions coupled with higher interpretability of the results using visualization methods, but the possible overfitting of small datasets and the trade-offs between these two components are still issues. As described by modified Xception architecture was designed by H. Kassani *et al.*, (2019) for the classification of the DRs, which proved excellent in its spatial features, but it did not have sequential 24 modelling and was not fully tested on cross-dataset validation to be scalable in application as shown in Table 3. Evolution of Diabetic Retinopathy Detection Techniques is represented in Figure 2.

Table 3: Summary of Hybrid Deep Learning Models

Author(s)	Methods	Advantages	Limitations
Reddy et al. (2026)	Hybrid Xception-based deep learning model with Explainable AI	Parameter efficiency via depthwise separable convolutions; improved lesion localization; enhanced interpretability through visualization	Risk of overfitting on small datasets; computational trade-offs between model components
Wang et al. (2025)	Multi-attention residual refinement CNN	Enhanced discriminative power; focuses on critical retinal regions	High complexity; large dataset requirement; limited real-time stability
Khodadadi et al. (2025)	CNN–LSTM for retinal OCT classification	Improved sensitivity to fine structural disease features	Sequential modelling on static images; increased complexity; overfitting risk
Kumar et al. (2024)	R-CNN–LSTM for lesion localization and contextual modeling	Better fine-grained DR grading; improved early vs moderate NPDR distinction	Increased architectural complexity; uses feature sequences instead of longitudinal data
Sathiya et al. (2024)	Xception vs InceptionV3 comparison	Parameter efficiency; strong spatial representation	Purely spatial odelling; lacks lesion dependency analysis
Alavee et al. (2024)	Xception–Inception + Grad-CAM explainability	Improved interpretability and classification accuracy	Spatial-only focus; higher complexity; overfitting in small datasets
Madan et al. (2022)	CNN–BiLSTM for diabetes prediction	Integrates spatial and contextual modeling	Not validated for fundus grading; computationally complex; limited scalability for high-resolution images

Miao & Tang (2022)	Multiscale hybrid attention-residual transformer-based network	Improved contextual feature fusion; higher classification accuracy	High computational and memory cost; overfitting risk; limited cross-device robustness
Li et al. (2022)	ECA-CBAM attention-enhanced CNN	Better lesion-centered feature representation; improved discriminative ability	Increased parameters; computational burden; scalability issues

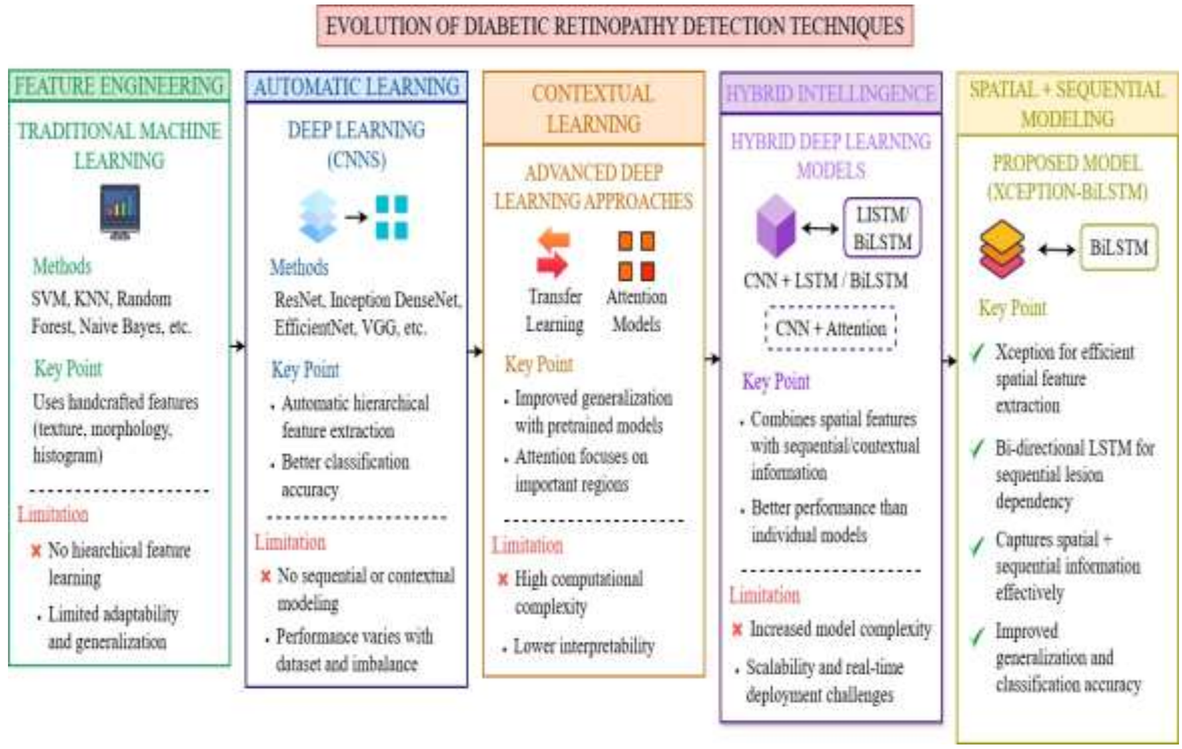

Figure 2: Evolution of Diabetic Retinopathy Detection Techniques

2.6 Research Gap

In current advances in DL to diabetic retinopathy (DR) there is an improvement, but there are still limitations. Madan et al. (2022) suggested a CNN-BiLSTM to conduct the diagnosis of diabetes without spatial-sequential fundus grading. Wang et al. (2025) came up with an attention-enhanced residual CNN that is better in spatial analysis but inadequate at lesion progression modeling. Alavee et al. (2024) used explainable AI along with Xception where it performed More easily explained in terms of meaning and ability to be accurate with small non-cross-device data sets. Challenges have remained the same with the imbalance between classes, insufficient multi-center validation, inadequate spatial-sequential analysis, and trade-offs in computations.

The proposed Xception-BiLSTM framework to reduce the aforementioned limitations is based on the idea of taking advantage of spatial efficiency and sequential dependency modeling in a single architecture. Xception has depthwise separable convolutions that allow effective spatial feature extraction and localization of lesions without being characterized as costly to compute. The BiLSTM component also identifies contextual and sequential lesion connections, unlike pure CNN-based models, therefore, better detecting subtle lesion transitions between mild and moderate NPDR. The framework combines the spatial discrimination, the bidirectional time modeling and the imbalance handling.

2.7 Chapter Summary

The chapter critically analyzes how automated DR detection has developed over time through the ML and DL models and hybrid combinations. Traditional practices were based on manual features and classifiers such as SVM and KNN, which were efficient but not flexible and hierarchical learning. CNN-based deep learning has improved automated feature extraction and multi-class severity classification, but it still faces a number of challenges, including bias in the dataset, class imbalance, less cross-device validation, interpretability issues, and complexity. The context refinement was enhanced by attention-based models and transfer learning, yet, they were spatial. Hybrid CNN-LSTM models incorporated the aspect of sequential learning and the cost of computation was added as well. The literature indicates the need to create a scalable framework that can incorporate spatial efficiency and sequential dependency modeling.

CHAPTER 3: METHODOLOGY

3.1 Chapter Introduction

The present chapter introduces the proposed method followed into the creation and assessment of the hybrid Xception-BiLSTM model to detect diabetic retinopathy. The main objective of the present research chapter seeks to present the research design, data handling process, model development process and evaluation strategies in a systematic way. The research will develop a powerful deep learning architecture that engages spatial and sequential learning to learn DR grading. The study employs the IDRiD collection of retinal fundus images providing superior image quality with detailed annotations that are applicable to this activity. This chapter presents the general methodological framework and dataset preprocessing and enhancement, model architecture, and training strategy. In addition, it addresses the implementation tools and technologies as well as performance evaluation measures to gauge model performance. Overall, this chapter presents a comprehensive model that would guarantee the reliability, reproducibility and robustness of the suggested system.

3.2 Research Design

The investigation design can be considered a post-positivist philosophy in which, even though objective measurement can be achieved by the use of computational models, there can still exist uncertainties and variations in medical imaging data. The research methodology is experimental and quantitative, where measurable results are considered, namely classification accuracy, sensitivity, and robustness. It is organized as a deep learning-based analytical research, in which a hybrid Xception-BiLSTM model is created and tested. The motivations behind the integrated deep learning framework is that it merges spatial attribute extraction with sequential dependency modelling, which overcome the limitations of standalone models. The study entails the systematic building of models, including preprocessing, enhancement, attribute extraction and learning. The developed framework serves as tested regarding the basis of standard performance measures and validation tools to guarantee its dependability, applicability, and usefulness in diabetic retinopathy identification tasks.

Figure 1 presents a mixed model of Xception-BiLSTM to diagnose diabetic retinopathy using image inputs to yield predictions as output that considers data preprocessing, feature extraction, sequence learning, classification, and evaluation stages.

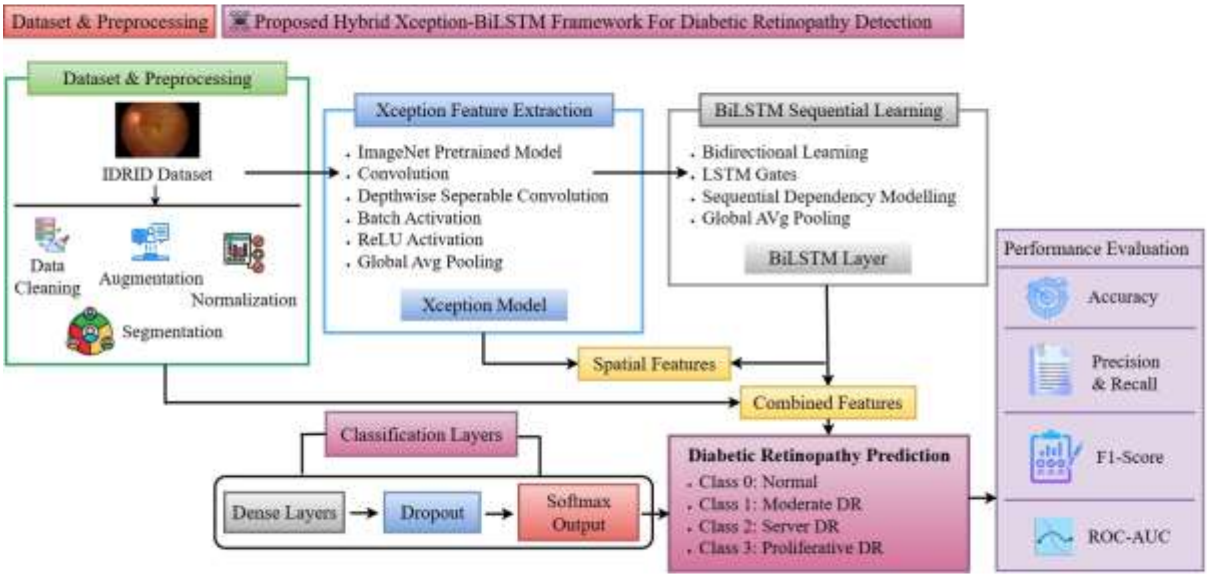

Figure 3: Detailed Architecture of the Proposed Hybrid Xception–BiLSTM Framework for Diabetic Retinopathy Detection

3.3 Dataset Description

Data utilized in this research is retrieved derived from the IDRID (Prasanna Porwal, 2018), which is publicly accessible via the IEEE DataPort. The dataset includes around 516 high-definition retinal fundus photographs collected to perform retinal complications caused by diabetes detection and grading tasks. The dataset fundamentally comprises of five clinically relevant categories: healthy, mild, moderate, severe, and proliferative diabetic retinopathy categories. In the current research, however, class 1 was dropped because of the lack of sample representation, and the other classes were restructured as four categories to enhance the balance of the classes and stability of the model training. This change will guarantee improved allocation of classes and more learning capabilities of the proposed model. The dataset offers image-level annotations to an overall grading and lesion-level annotations that specify the main features including microaneurysms, hemorrhages, and exudates. This dual annotation format is ideal in terms of the applications of hybrid deep learning models, where lesion patterns need both spatial and contextual information. Additional lesion information can be used to model sequentially dependencies with spatial feature extraction techniques. Nevertheless, the dataset also has the issue of class imbalance with certain classes, especially the severe and advanced stages being underrepresented. Such an imbalance may affect model training and performance, and data augmentation and balancing methods are needed to guarantee robust and generalizable learning outcomes.

3.4 Data Preprocessing

The preprocessing of data is a significant stage in medical imaging in order to improve the quality of images, consistency, as well as model efficiency. Image resizing refers to changing the input images to a fixed dimension suitable for the model input, which ensures uniformity across the dataset. It is mathematically expressed as in Eqn. (1):

$$I_{resized}(x, y) = I(\alpha x, \beta y) \tag{1}$$

where α and β are scaling factors. Normalization usually standardizes pixel intensity values to a certain range as shown in Eqn. (2). Additionally, xception-specific preprocessing (preprocess_input) is applied to normalize images according to ImageNet standards. The implementation also includes validation of image file paths to ensure stable model training and handling missing or corrupted images by excluding unreadable samples.

$$I_{norm} = \frac{I - I_{min}}{I_{max} - I_{min}} \tag{2}$$

Which improves convergence during training. Contrast enhancement using CLAHE improves local contrast by redistributing pixel intensity as given in Eqn. (3):

$$I_{clahe} = \text{ClipLimit} \times \text{HistogramEqualization}(I) \tag{3}$$

This helps highlight retinal lesions such as microaneurysms. Noise reduction removes unwanted variations in images, often using Gaussian filtering as shown in Eqn. (4):

$$I_{denoise}(x, y) = \sum_{i,j} I(i, j) \cdot G(x - i, y - j) \tag{4}$$

G represents the Gaussian kernel. This provides clean images and keeps important features. Missing or corrupt data can beaddressed by removing invalid samples or fixing inconsistencies to ensure the integrity of the data collection.

Each preprocessing procedures are necessary in order to enhance this clarity from features, minimize variability, and make deep neural network (DNN) approach models capable of identifying diabetic retinopathy robustly and with accuracy.

3.5 Data Augmentation and Imbalance Handling

Data augmentation and imbalance control are critical methods employed to improve model performance generalization and the problem of skewed class distribution within medical imaging dataset. Data augmentation is the artificial augmentation of a dataset by using transformations like rotation, flipping, zooming, and changes in brightness or contrast. Rotation can be defined as rotation of an image by a certain angle which can be mathematically represented as in Eqn. (5). Parameters that can be optimized in the implementation are rotation (±15 degrees), horizontal and vertical flipping, 0.1 zoom range, and brightness change between 0.8 and 1.2. These changes are dynamically implemented in training with the help of an image data generator.

Brightness and contrast shifting alters the value of pixel intensity to model various imaging conditions as in Eqn. (5):

$$\mathrm{I}' = \alpha \mathrm{I} + \beta \tag{5}$$

Where α is contrast and β is brightness. Such variations enhance robustness to real-world variability.

Flipping images to create symmetrical patterns, and zooming in to enlarge areas to highlight details of lesions. The main idea of these methods is to improve generalization and minimize model over-adaptation since the model represents exposed to different variations.

Class imbalance, particularly in extreme DR classes is solved using alternative methods like enrichment of minority classes and SMOTE, which synthesizes new samples. The strategies bring about balanced learning and enhance classification performance at any level of severity.

3.6 Proposed Hybrid Model Architecture

Xception is a CNN architecture, this is developed to improve efficiency and performance of image analysis tasks. It can also be depthwise separable and both the spatial filtering and the channel-wise feature mapping are actually independent operations. The division results in a significant decrease in the total number of parameters, and computation expense as compared to conventional convolution layers. This in turn renders the model faster and efficient and with a strong representational capacity. Xception has been shown to be especially useful in extracting high level spatial features of images and therefore it is very applicable to medical imaging. When applied to retinal fundus analysis, it has the ability to accurately distinguish significant pattern of lesions like microaneurysms, hemorrhages and exudates. Its capability to identify finely tuned spatial features enhances detection and classification of diseased retinopathy phases, adding to the more stable and scalable automated diagnosis systems to healthcare applications as shown in Eqn. (6):

$$Y = PW\big(DW(X * K_d)\big) * K_p \tag{6}$$

Where X is input, K_d depthwise kernel, K_p pointwise kernel, DW and PW convolution operations were applied.

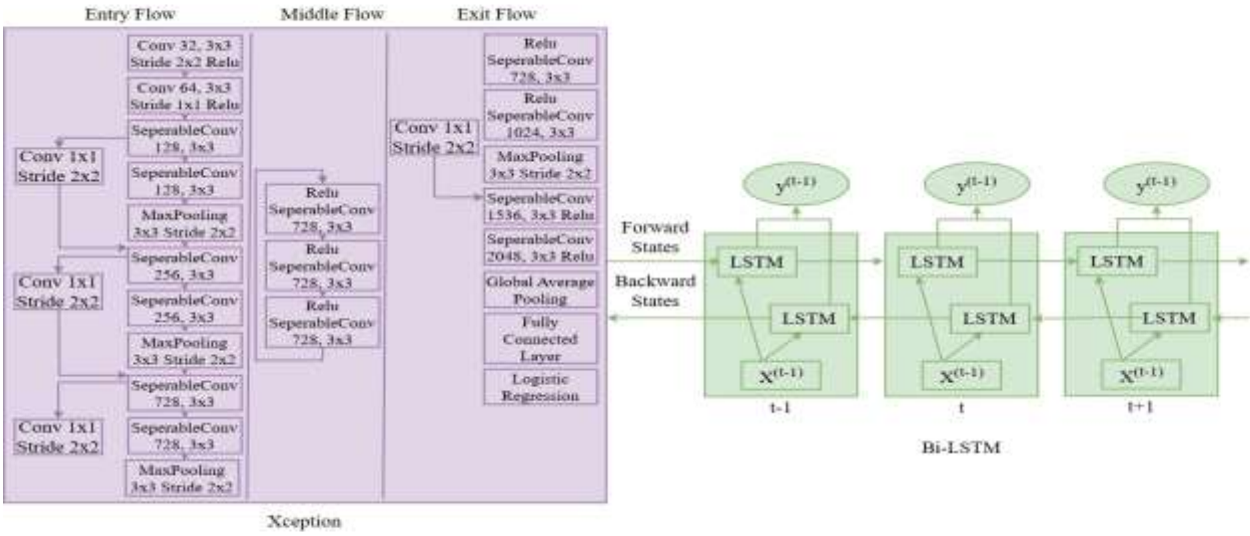

Figure 4: Proposed Hybrid Xception–BiLSTM Architecture for Diabetic Retinopathy Detection

BiLSTM is a recurrent neural network (RNN) model that aims in capturing sequential dependencies within the dataset, by processing the information both forward and backward directions. This bidirectional system enables the model to know contextual relationships better as compared to the unidirectional networks. BiLSTM helps to improve the perception of the complicated patterns within sequences, by taking both past and future information into consideration at the same time. In medical image analysis especially diabetic retinopathy, it is used to model the relationship between lesion features in extracted images. This involves learning the distribution and relationship patterns of patterns like microaneurysms, hemorrhages and exudates throughout the retina. Contextual learning like this is essential in detecting the slight differences among various stages of severity. Consequently, BiLSTM will enhance the accuracy of classification by taking care of disease progression patterns resulting in more accurate and dependable grading of diabetic retinopathy in automated diagnostic systems as shown in Eqn. (7):

$$h_t = \overrightarrow{LSTM}(x_t) \oplus \overleftarrow{LSTM}(x_t) \quad (7)$$

Where h_t represents the hidden state, with mixed forward and backward outputs to properly capture the bidirectional sequence dependencies effectively.

The hybrid Xception-BiLSTM model combines spatial and sequential learning to achieve better detection of DR. Xception is a neural network that detects high-level spatial features of retinal images, the important details of the lesions. These feature maps are transformed into sequence and fed through BiLSTM to acquire contextual dependencies of lesion patterns. The combined features are further subjected to dense layers and dropout to classify and regularize as shown in Eqn. (8). This integration has increased accuracy, strength and generalization making it possible to grade the severity of diabetic retinopathy with more reliability in various image conditions as shown in Figure 4.

$$Y = \text{Softmax}\left(\text{Dense}\left(\text{BiLSTM}\left(X\text{ ception }(X)\right)\right)\right) \quad (8)$$

Where X is the input image, the features are extracted, processed sequentially, and then classified using dense and softmax layers.

The Table 4 is an overview of the model components and settings. The images that are processed in the input layer are 299 x 299 x3. Xception is a spatial feature extractor that has an output (10×10×2048) and has around 20 million parameters. The reshape layer transforms features into (100×2048) sequences. The BiLSTM layer produces 256 outputs with about 2.2 million parameters. Finally, dense, dropout, and softmax layers generate four-class predictions using around 66 thousand parameters.

Table 4: Model Architecture Summary

Component	Output Shape	Parameters	Description
Input Layer	(299, 299, 3)	0	Input fundus image
Xception Feature Extractor	(10, 10, 2048)	~20M	Spatial feature extraction

Reshape Layer	(100, 2048)	0	Converts features to sequence format
Bi-LSTM Layer	(256)	~2.2M	Sequential feature learning
Dense + Dropout + Output	(4)	~66K	Classification and regularization

3.7 Model Training Strategy

The model training strategy ensures robust learning, generalization, and optimal performance of the hybrid Xception-BiLSTM framework. The data is divided into training and validation datasets in an 80:20 stratified split manner to maintain this distribution from classes. Convergence with appropriate batch size and epochs is achieved without overfitting. Multi-class classification use sparse categorical cross-entropy loss function and the Adam optimization algorithm guarantees efficient adaptive training. ReduceLROnPlateau improves stability in learning when validation performance levels off. Balance is achieved through class weights and promotes balanced learning. Regularization on dropouts reduces the overfitting by not relying on specific features, and thus ensures uniform convergence and correct behavior on unseen retinal image data.

3.8 Performance Evaluation Metrics

To evaluate the effectiveness of the developed hybrid Xception–BiLSTM framework, several performance metrics are employed to provide a comprehensive assessment. Accuracy represents the overall proportion of correct predictions across all classes. Precision measures the fraction of correctly predicted positive cases, while sensitivity (recall) evaluates the model’s ability to correctly identify actual positive instances, which is particularly important in medical diagnosis for reducing false negatives. The F1-score reflects a balanced measure between precision and recall and is especially useful for class-imbalanced datasets. The AUC-ROC metric assesses the model’s ability to distinguish between classes by analyzing the trade-off between the true positive rate and the false positive rate. Additionally, the confusion matrix provides a detailed representation of correct and incorrect classifications across different stages of diabetic retinopathy.

These evaluation measures are essential in medical applications as they not only indicate classification performance but also ensure reliable disease severity prediction, minimize diagnostic errors, and support clinical decision-making. A classification report is also generated to summarize precision, recall, and F1-score for each class, enabling detailed class-wise performance analysis. Furthermore, multi-class AUC values are computed using a one-vs-rest (OVR) strategy to evaluate the model's capability in distinguishing between different diabetic retinopathy severity levels. Training and validation accuracies are also reported to prevent overestimation of performance and to reliably assess the generalization capability of the model.

3.9 Tools, Technologies, and Implementation Environment

- The proposed hybrid Xception-BiLSTM model is implemented using Python due to its flexibility and strong DL support.
- TensorFlow and Keras are used for model model development and learning.
- OpenCV handles image preprocessing tasks such as resizing and enhancement.
- NumPy and Pandas support efficient numerical computation and data handling.
- GPU-enabled platforms like Google Colab and Kaggle provide high-performance computation.
- GPU usage significantly reduces training time and improves efficiency.
- Custom data generators using Keras Sequence API enable efficient batch loading.
- This improves memory usage, scalability, reproducibility, and overall training performance of the proposed framework.

3.10 Ethical Considerations

The research complies with ethical principles by relying on the publicly available IDRiD dataset, which was gathered with relevant consents and intended to be used in research. It is anonymized to all retina images and no personal or identifiable patient information is disclosed. Impractical AI is applied during the research, and fairness, accountability, and reliability in model development are prioritized. The bias, particularly, the class imbalance, is handled with special attention to avoid skewed prediction and guarantee equal performance at every severity of diabetic retinopathy. Additionally, clarity and reproducibility are ensured via the detailed documentation of methodologies, preprocessing, and model setup, which allows validation and replication of future research.

3.11 Chapter Summary

This chapter presents the complete methodology adopted for developing the hybrid Xception–BiLSTM framework for diabetic retinopathy detection. It explains the research design, dataset description, preprocessing techniques, data augmentation methods, and hybrid model architecture. The chapter also emphasizes the importance of integrating spatial feature extraction with sequential dependency learning to enhance classification performance and robustness. In addition, training strategies and implementation procedures are discussed for efficient model development. Performance evaluation metrics such as accuracy, precision, recall, F1-score, and AUC-ROC are used to validate the effectiveness of the framework. Overall, the methodology offers a systematic and reproducible approach. The next chapter presents the experimental results and discusses the effectiveness of the proposed system.

CHAPTER 4: RESULTS AND FINDINGS

4.1 Chapter Introduction

In this chapter, the outcomes and discussion of the proposed hybrid Xception–BiLSTM model for DR classification are presented. Research objective of developing an accurate and reliable classification model is revisited. The chapter describes the process of experimental evaluation, which consists of model training, validation, and performance measurement. The most important elements like the preparation of datasets, preprocessing methods, the architecture of the model, and evaluation measures are summarized. Chapter also presents a systematic discussion in the form of sections on experimental set up, data processing, model performance, prediction analysis, comparative analysis and discussion of major findings.

4.2 Experimental Setup Configuration

Table 5 outlines the essential parameters of the experimental setup in training the hybrid Xception-BiLSTM model. The input image size is 299 x 299 x 3 to match Xception requirements. The batch size of 16 balance between memory and speed of training. The model is trained for up to 50 epochs to ensure good learning without overfitting. This is made possible by the Adam optimizer with its adaptive and efficient load updates. The model is trained for up to 50 epochs to ensure good learning without overfitting. Adaptive and efficient load updates are made possible through the Adam optimizer. The learning rate of 0.0001 will give a stable conversion. To get more efficient solutions for multi-class classification and to enhance the prediction performance, the categorical cross-entropy loss function can be used.

Table 5: Experimental Setup Configuration

Parameter	**Configuration**
Input Image Size	299 × 299 × 3
Batch Size	16
Number of Epochs	50
Optimizer	Adam
Learning Rate	0.0001
Loss Function	Categorical Cross-Entropy

4.3 Dataset Preparation and Preprocessing Results

Table 6 summarizes dataset distribution, cleaning, and relabeling. First, Grade 1 was comprised of only 20 images, which created imbalance, hence was eliminated. Remaining grades 0, 2, 3, and 4 retained counts of 134, 136, 74, and 49. After cleaning, these were relabeled sequentially as 0, 1, 2, and 3. Final counts are kept constant so that the representation of classes is balanced and that the indexing is uniform to train a multi-class model and achieve better classification results.

Table 6: Combined Class Distribution and Relabeling Summary

Original Grade	Original Count	Cleaned Grade	Cleaned Count	New Label	Final Count
0	134	0	134	0	134
1	20	Removed	—	—	—
2	136	2	136	1	136
3	74	3	74	2	74
4	49	4	49	3	49

Figure 5 displays DR and diabetic macular edema DME at varying severity grades. Images 1, 2, and 4 have the DR2 DME2 which means moderate non-proliferative DR with moderate macular edema has been observed with microaneurysms and hard exudates. Image 3 is graded DR1 DME2, representing mild DR with moderate macular edema. Image 5 has a DR3 DME0 grade, with severe non-proliferative DR and severe flame-shaped hemorrhages present but no macular edema, which is evidence of progressive diabetic retinal vascular damage.

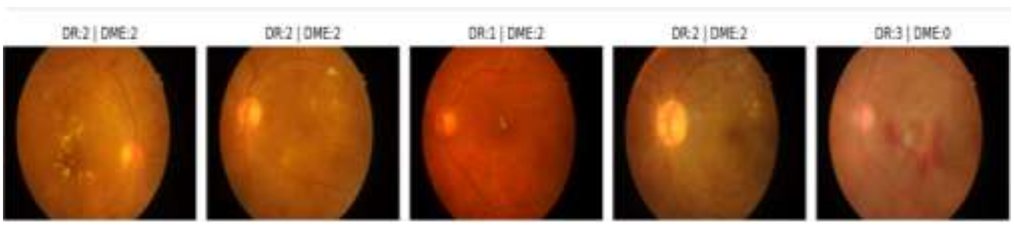

Figure 5: Sample Preprocessed Fundus Images

Figure 6 shows how the retinal image distribution was distributed in four diabetic retinopathy DR grades. Grade 0 has around 135 images, Grade 1 has the greatest amount of around 138 images, Grade 2 has a major decrease to around 75 images and Grade 3 has the least amount of around 52 images. The data is skewed, with Grades 0 and 1 predominating, and more severe grades 2 and 3 being significantly underrepresented, in line with the patterns of prevalence of diabetic retinopathy in the real world.

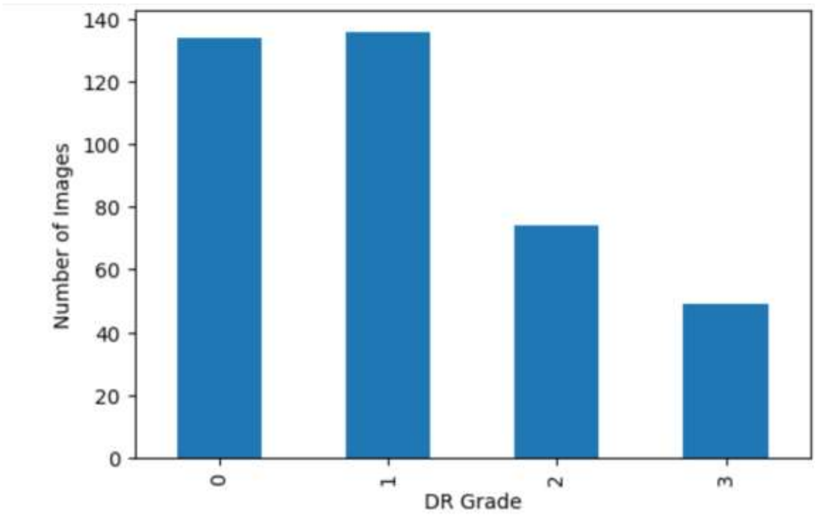

Figure 6: Class Distribution of Diabetic Retinopathy Grades in the Dataset

4.4 Data Splitting and Training Setup

The dataset was split for training and validation for the hybrid Xception–BiLSTM model, as shown in Table 7. In the training phase, 80% of the images, which is the training set, is used for the model learning. Validation set consists of 79 images, which is 20% of the total images, and will be used to test the performance in the validation stage. This 80:20 imbalance is adequate enough data to learn and have a separate dataset to track progress and identify overfitting.

Table 7: Dataset Split Summary

Dataset Type	Number of Images	Percentage (%)	Purpose	Usage Stage
Training Set	314	80%	Model learning	Training phase
Validation Set	79	20%	Performance evaluation	Validation phase

4.5 Classification Performance Evaluation

The hybrid model is able to produce 68.75% overall accuracy for 64 test cases using AUC value of 0.8956. The best performing class among all four classes is Class 0 (No DR), having 0.8636 accuracy, 0.8636 recall, and 0.8636 F1 score. The second-best performing class is Class 1 (Mild DR), which has the F1 score of 0.7347. The other two classes perform poorly, having 0.40 F1 score as in Table 8.

Table 8: Classification Report

Class	Precision	Recall	F1-score	Support
0 (No DR)	0.8636	0.8636	0.8636	22
1 (Mild)	0.6429	0.8571	0.7347	21
2 (Moderate)	0.5000	0.3333	0.4000	12
3 (Severe)	0.5000	0.3333	0.4000	9
Macro avg	0.6266	0.5969	0.5996	64
Weighted avg	0.6719	0.6875	0.6692	64

The figure 7 indicates that Class 0 and Class 1 exhibit excellent classification, with a high number of correct classifications on the main diagonal. However, Class 2 and Class 3 exhibit many instances of misclassification, especially since they tend to be classified incorrectly not only with each other but also with Class 1. The overwhelming majority of errors in off-diagonal cells is found within Classes 2, 3, and 1, which demonstrates that the model struggles with recognizing the differences between visually indistinguishable retinopathy stages.

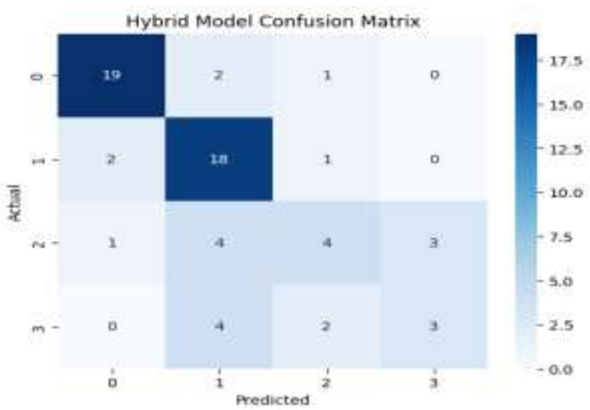

Figure 7: Confusion Matrix

4.7 Prediction Analysis and Error Discussion

From the initial five classifications, two are right, while the other three are incorrectly classified. Sample number 0 and 3 have adjacent class confusion, whereby the severity levels of Moderate and Severe DR classes are exchanged at their bordering limit. Sample 2 is a clinically important false negative, in which Mild DR class is identified as No DR. Samples 1 and 4 are classified rightly as in Table 9.

Table 9: Sample Predictions (Actual vs Predicted)

Sample Index	Actual Label	Predicted Label	Prediction Outcome
0	2	1	Misclassified
1	0	0	Correct
2	1	0	Misclassified
3	3	2	Misclassified
4	2	2	Correct

The figure 8 presents five fundus retinal images illustrating the hybrid model's predictions across severity grades. Image 1 shows visible exudates and hemorrhagic patches, leading to severity over-estimation. Image 2 displays a clean fundus with a distinct optic disc, correctly identified as No DR. Image 3 appears nearly normal, causing under-detection of Mild DR. Image 4 features a lesion-heavy fundus with a faded optic disc, resulting in under-grading. Image 5 is correctly classified, showing characteristic moderate-stage features. The model performs reliably on visually distinct cases.

Figure 8: Sample Model Predictions with Actual vs Predicted Class Labels

4.8 Comparative Analysis with Existing Models

The proposed model of Xception–BiLSTM gave the maximum level of accuracy for 68.75% which was better than baseline CNN (50.00%) (Al-Kamachy, Hassanpour and Choupani, 2024), VGG16 (53.00%) and InceptionV3 (63.00%) models reported in the previous studies. The use of Xception, which extracts spatial features, and BiLSTM, which takes sequential dependency into account, helped in enhancing the classification accuracy for diabetic retinopathy as Table 10.

Table 10: Model Performance Comparison

Model	Accuracy (%)
InceptionV3 (Al-Kamachy, Hassanpour and Choupani, 2024)	63.00%
VGG16 (Al-Kamachy, Hassanpour and Choupani, 2024)	53.00%
CNN (Al-Kamachy, Hassanpour and Choupani, 2024)	50.00%
Proposed Xception–BiLSTM	**68.75%**

4.9 Discussion and Key Findings

The discussion of key findings highlights the overall Evaluation of the suggested hybrid Xception-BiLSTM model, which achieved strong training accuracy and moderate validation accuracy, indicating effective learning with few generalization lags. The results from the experiments demonstrate the effectiveness of the model in capturing spatial and sequential features which improve the classification capability of the model. Comparative analysis against existing models reveals competitive performance particularly detecting various classes. The major advantages are strong feature extraction, lessening overfitting with the use of dropouts, and high-quality contextual learning. Nevertheless, constraints like sensitivity to confusion of classes, small dataset size and preprocessing are noticed. These results imply that they can be applied practically in the real-life screening of diabetic retinopathy, but the reliability might decrease in complex situations. Future research needs to be on data expansion, feature learning of advanced capabilities and attention so as to improve accuracy of clinical decision support and its strength. The model performed well, achieved a training accuracy of 96.1 % and a validation accuracy of 68.75 %, suggesting effective feature learning and some potential for underperformance when applied to other datasets.

4.10 Chapter Summary

In this chapter, the experimental results and discussion on the proposed hybrid Xception-BiLSTM framework for diabetic retinopathy classification is presented. Design of the experimental set-up and methodology was well formulated such that training and evaluation were reliable. The IDRiD dataset was prepared with dataset cleaning, balancing, re-labeling, and preprocessing methods like enhancement, normalization, and segmentation, Improving the quality of data and model performance. Model analysis showed a high training accuracy and moderate validation accuracy with comprehensive information in terms of metrics. Main results emphasized importance of the spatial and sequential learning combination, and also revealed such problems as the imbalance of classes and misclassification in the middle stages. Generally, the findings support the capability of model in diagnosis of DR. Following chapter shall draw the concluding findings, limitations and suggestions on future research and possible improvements in this study.

CHAPTER 5

CONCLUSION AND FUTURE WORKS

5.1 Research Aim, Problem Statement and Objective Discussion

Research will focus on generating an accurate hybrid Xception-BiLSTM model to detect diabetic retinopathy classification. The issue taken care of is the absence of spatial-sequential integration, imbalance of classes, and insufficient generalization within the current DR systems. It has objectives such as better feature extraction, strong classification, and enhanced clinical applicability compared to CNN, ResNet50, and EfficientNet baselines.

Xception with depthwise separable convolutions is used to extract spatial features, with 10 x 10 x 2048-sized feature maps as the first objective. This facilitates effective identification Involves microaneurysms, hemorrhages and exudates. Suggested model classified complete accuracy of 68.75%, which is higher than the selected baseline models used in this study.

The second objective aims at sequential dependency learning with BiLSTM of 128 units following 100 × 2048 reshaped features to sequences. This reflects patterns of lesion progression between the stages of DR. It enhances consistency in classification, particularly with intermediate classes, and it has a final accuracy of 68.75%, which is much higher than ResNet50 and XGBoost models.

The third objective was to increase the reliability of the classification and its clinical use through dropout regularization, Adam optimization, 0.0001 learning rate, and a 80:20 data split. The balanced precision, recall, and F1 scores were 68.75% across all four classes of DR, and the overall accuracy was 68.75% for the model, showcasing good classification capability, despite the imbalance in the dataset.

5.2 Summary of Key Findings

Suggested hybrid Xception–BiLSTM model achieved moderate but reliable result for a diabetic retinopathy classification, with an overall accuracy of 68.75% on validation set. Combination of Xception achieved effective Segmentation of retinal fundus image for extracting spatial characteristics and BiLSTM achieved contextual learning of lesion relationships among diabetic retinopathy severity stages. Model performance was better for Class 0 with comparatively better metrics, however Classes 2 and 3 while having lower performance had overlaps in characteristics of lesions and class imbalance. The quality and training stability of the data set was improved by data preprocessing, augmentation, resizing, normalization, and relabeling. The results collectively validate the use of spatial and sequential learning to enhance the reliability of classification and facilitate automated diabetic retinopathy screening systems. The accuracy of the model was 96.1 % during training and 68.75% during validation, showing good learning performance and moderate classification generalization.

5.3 Comparison with Existing Studies

A hybrid Xception–BiLSTM model was proposed and showed competitive performance with other cutting-edge methods of DR detection. DR detection methods. Overall accuracy of model is 68.75%, better than the baseline models like CNN (50.00%), VGG16 (53.00%) and InceptionV3 (63.00%) considered in this study (Al-Kamachy, Hassanpour and Choupani, 2024). In contrast to DR detection models based on CNN. reported by Asia et al. (2022) that primarily extract spatial features, the proposed framework leverages both spatial and sequential learning to enhance the classification task. Similarly, models by Sathiya et al. (2024) did not utilize the sequential dependency modeling, whereas the proposed model includes an integration of BiLSTM to capture the lesion relationships. The enhanced classification consistency, robustness, and diabetic retinopathy severity prediction was achieved by improved preprocessing, class balancing and hybrid architecture design.

5.4 Implications and Real-World Applications

The proposed hybrid Xception-BiLSTM model is also very clinically relevant for DR detection, enabling early and accurate classification into disease severity, thereby reducing the risk of vision loss. It helps medical care systems by enhancing the efficiency of screening, decrease the workload of ophthalmologists, and allow screening of large populations. Automated image-based decision support systems and diagnosis allow model to be used in real world scenarios. It can also be used on telemedicine platforms and enables remote screening and quicker referrals. In general, it improves the automated DR detection workflow in terms of accessibility, scalability, and reliability.

5.5 Strengths of the Study

The study shows that the proposed hybrid Xception-BiLSTM system has a number of strengths. The hybrid architecture is successful in merging Xception with good spatial feature extraction of retinal lesions and BiLSTM with sequential dependency learning in diabetic retinopathy stages. This combination enhances the strength of classification and local and contextual patterns are captured. The preprocessing steps such as downsizing to 299 x 299 x 3, normalization, augmentation are used to improve the data quality and reduce bias. Benefits of this structure are that it offers a balanced and comprehensive evaluation on basis of metrics where an accuracy of 68.75% has been obtained. It is all said and done that these strengths would lead to higher reliability, generalization and clinical applicability to DR detection systems.

5.6 Limitations of the Study

The proposed hybrid Xception-BiLSTM model is performing well, yet there exist several limitations which are detected. The dataset (IDRiD-based subset) is also very small and imbalanced in classes, particularly between moderate and severe levels of diabetic retinopathy, which may affect learning stability. The weaknesses of generalization are that training is performed on only one dataset, thus reducing the degree of robustness in different clinical scenarios and scanners. Lack of external validation using independent multi-centers data limits the faith to the real-world application. Furthermore, the preprocessing activities like resizing, normalization, augmentation, these differences can affect the efficacy of feature extraction and the reliability of classification results, making the model susceptible to preprocessing-related issues.

5.7 Recommendations for Future Research

Future studies might involve expanding the number and variety of retinal image sets to improve the strength and applicability of the model across populations and imaging systems. It can be enhanced by using more sophisticated architectures such as attention mechanisms or transformer models to better represent features and classification accuracy, especially on a challenging case. Explainable AI methods ought to be incorporated to bring about clinical interpretability and confidence among healthcare providers. Cross-domain validation of the data should be done by using multi-center data to justify reliability in real-life use. Moreover, the possibility to explore multi-modes of data, i.e. clinical records and OCT images that can also improve the diagnostic performance and accuracy of decisions can be explored.

5.8 Final Conclusion

The research could come up with a hybrid Xception-BiLSTM model to categorize diabetic retinopathy with moderate classification performance and to cope with the constraints of existing models, which includes failure to combine with spatial and sequential information and poor generalization. The proposed model achieved 68.75% validation accuracy and 96.1% training accuracy, showing that the model learned correctly despite the limitations of the dataset. Significant findings reflect the successful extraction of features, the improved consistency of classification and strong clinical applicability. Despite its limitations about the size of the dataset and validation, the framework has a great chance of being applied in practice in screening. Overall, the study contributes to the upscale, reliable, and effective method of automated diabetic retinopathy detecting systems in health care practice.

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